

Explaining Time-series Classifiers Through Posthoc XAI Methods Capturing Temporal Dependencies

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Expected Contributions

- **Rule-Based Explanation:** Mimics black-box models using interpretable surrogates like decision trees.
- **LOMATCE:** Local model-agnostic explanations for time series classifiers without segmentation.
- **L2GX:** Aggregates local explanations to form holistic, global insights.

Motivation

- Deep learning models perform well on time series classification tasks; however, they are often considered as **black boxes** [1].
- Existing XAI methods for **image** and **tabular data** may not be suitable for time series data due to its **temporal nature** [1,2].
- Many time series XAI approaches are **developer-focused** and rely on **latent layer activations**, while users need **higher-level, abstract explanations** [2].
- Feature importance methods like **LIME**[3] applied to time series often segment data into **fixed-size windows**, potentially overlooking critical subsequences [2].

How can model-agnostic XAI methods be designed to provide interpretable, faithful explanations while capturing temporal dynamics in the data for black-box time-series classifiers?

Research Plan and Methodology

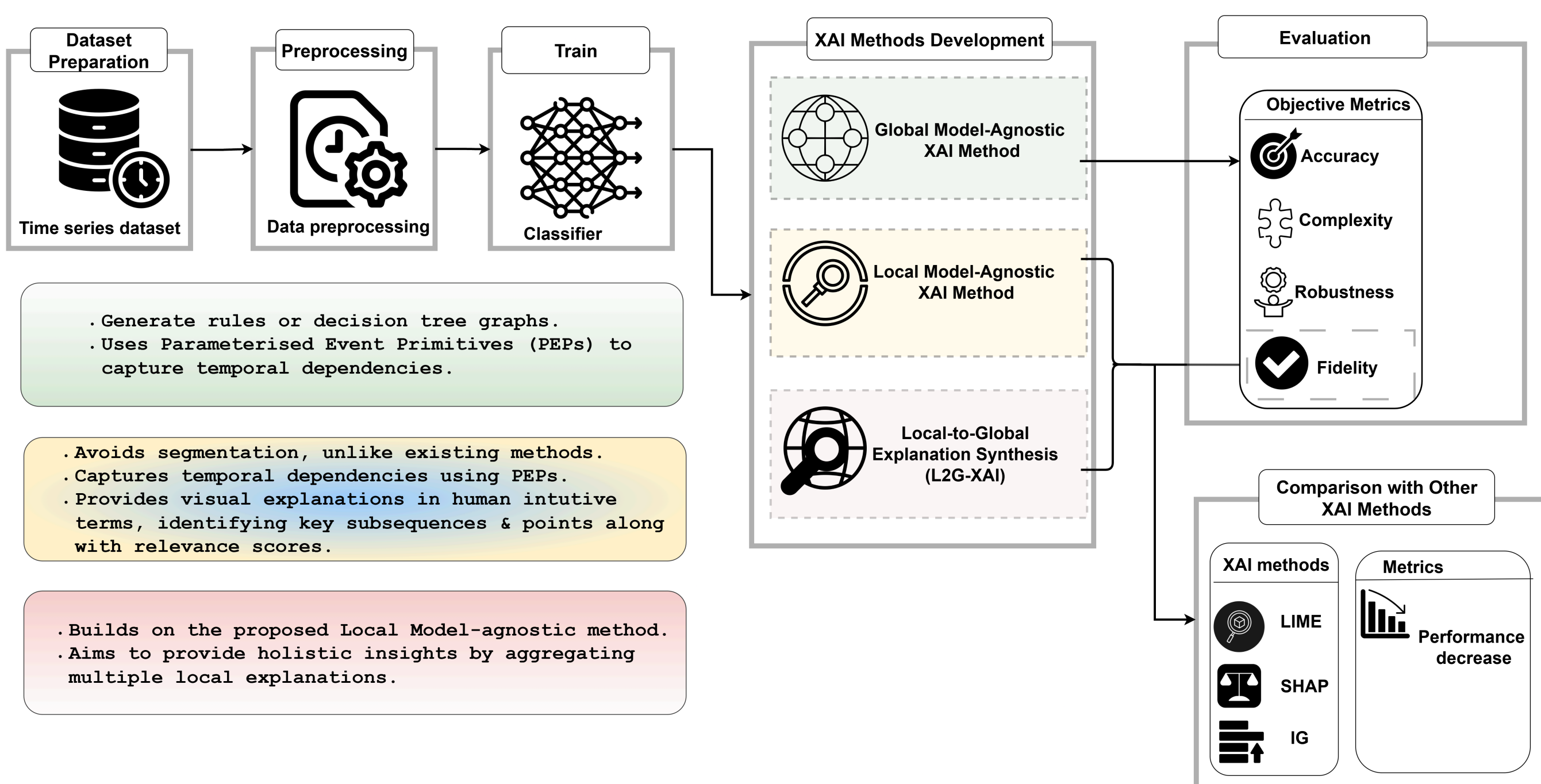


Fig 1: Overview of the proposed research plan.

Global Model-Agnostic Rule-Based XAI Method

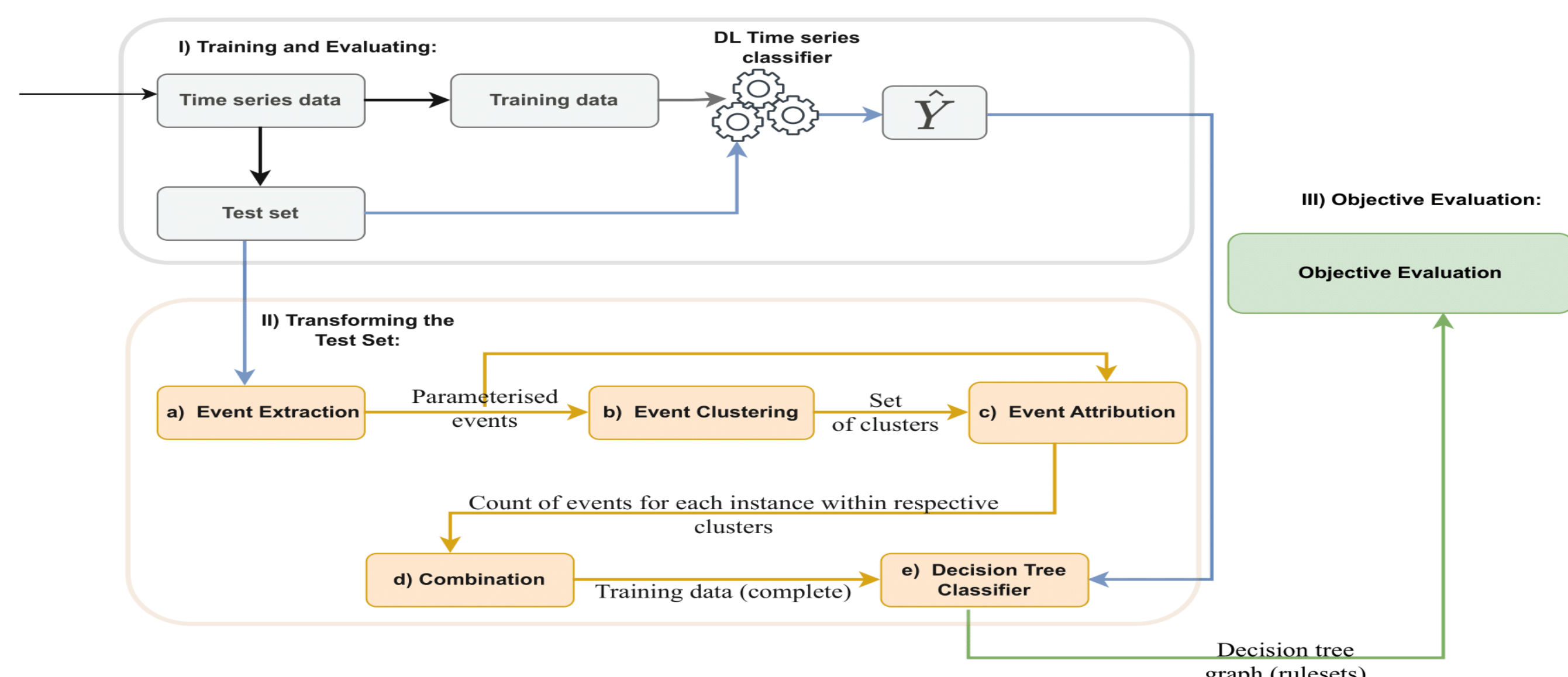


Fig 2: The proposed global XAI method for DL time series classification models.

LOMATCE: Local Model-Agnostic Time-series Classification Explanations

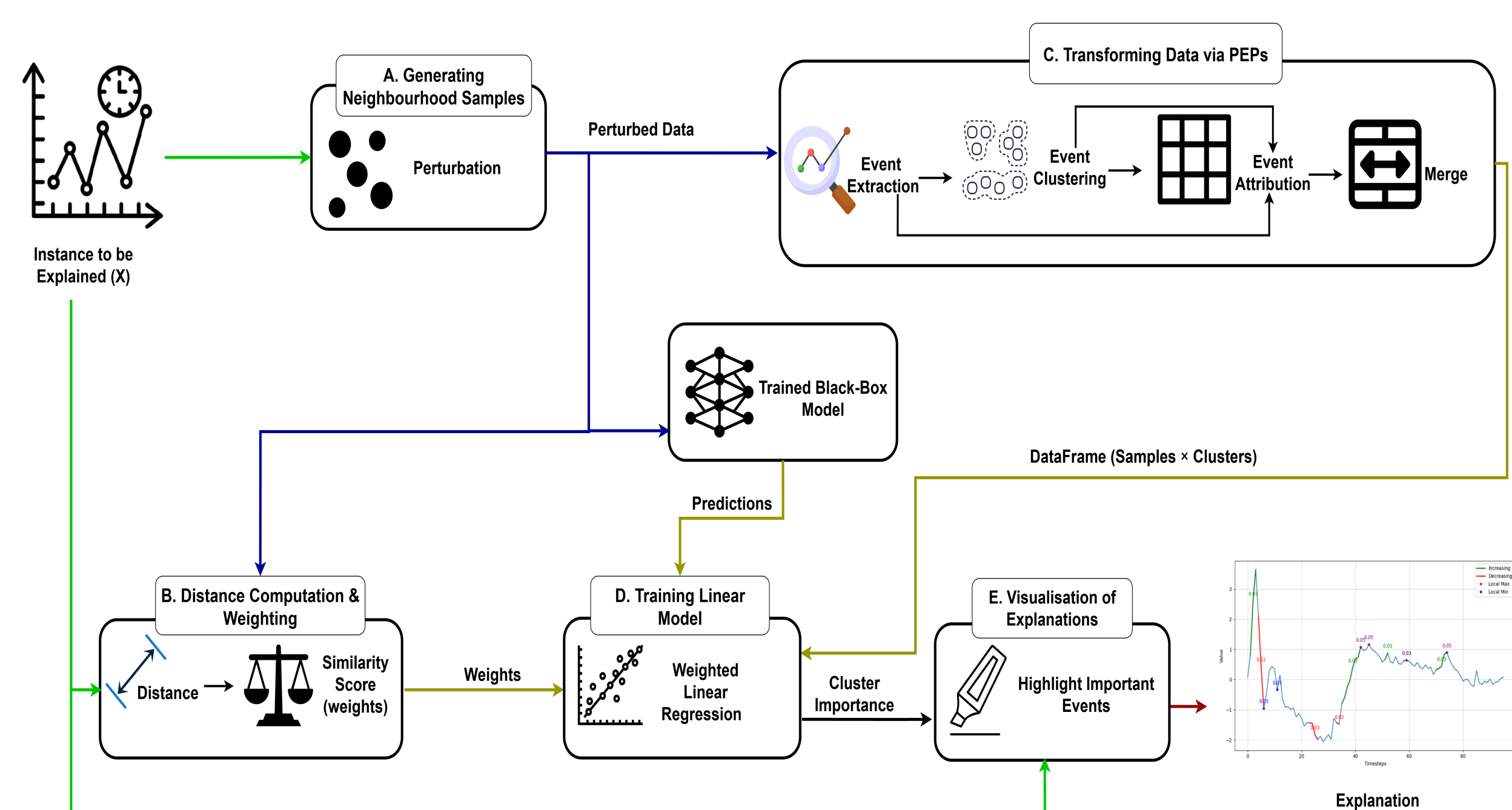


Fig 3: Step-by-step illustration of the proposed local XAI approach.

Result and Discussion

Global Model-Agnostic Rule-Based XAI Method

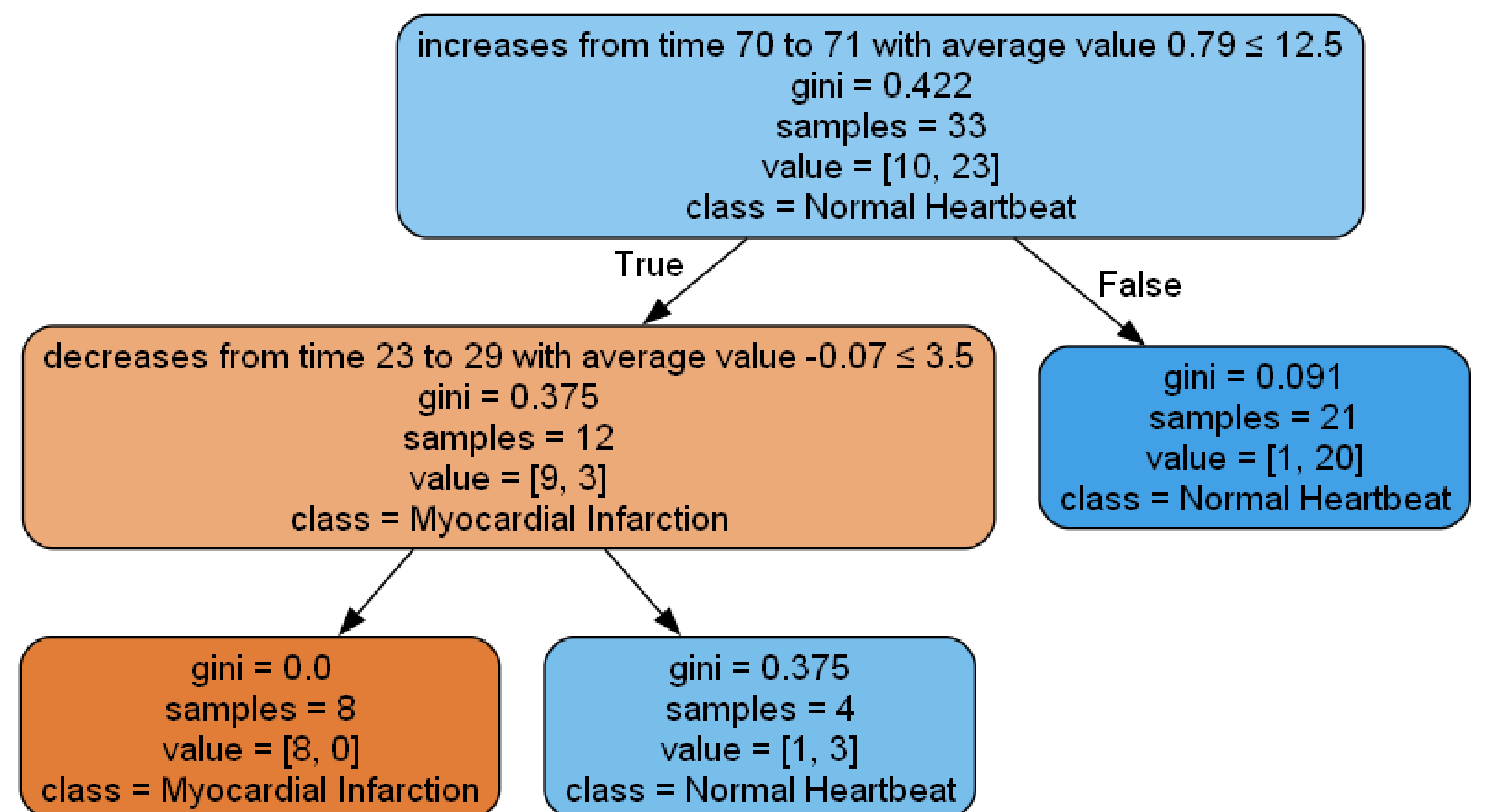


Fig 4: A decision tree graph produced by the proposed method on ECG data.

- The graph provides valuable insights into the impact of specific time steps and corresponding events on the model's predictions.

LOMATCE: Local Model-Agnostic Time-series Classification Explanations

Table 1: Explanation faithfulness, with 95% confidence interval, across various perturbation methods.

Dataset	Zero	Mean	Total_mean	Random
ECG	0.82 ± 0.02	0.70 ± 0.02	0.81 ± 0.01	0.29 ± 0.02
GunPoint	0.72 ± 0.02	0.52 ± 0.03	0.73 ± 0.02	0.32 ± 0.04
Coffee	0.65 ± 0.03	0.57 ± 0.02	0.66 ± 0.03	0.22 ± 0.08

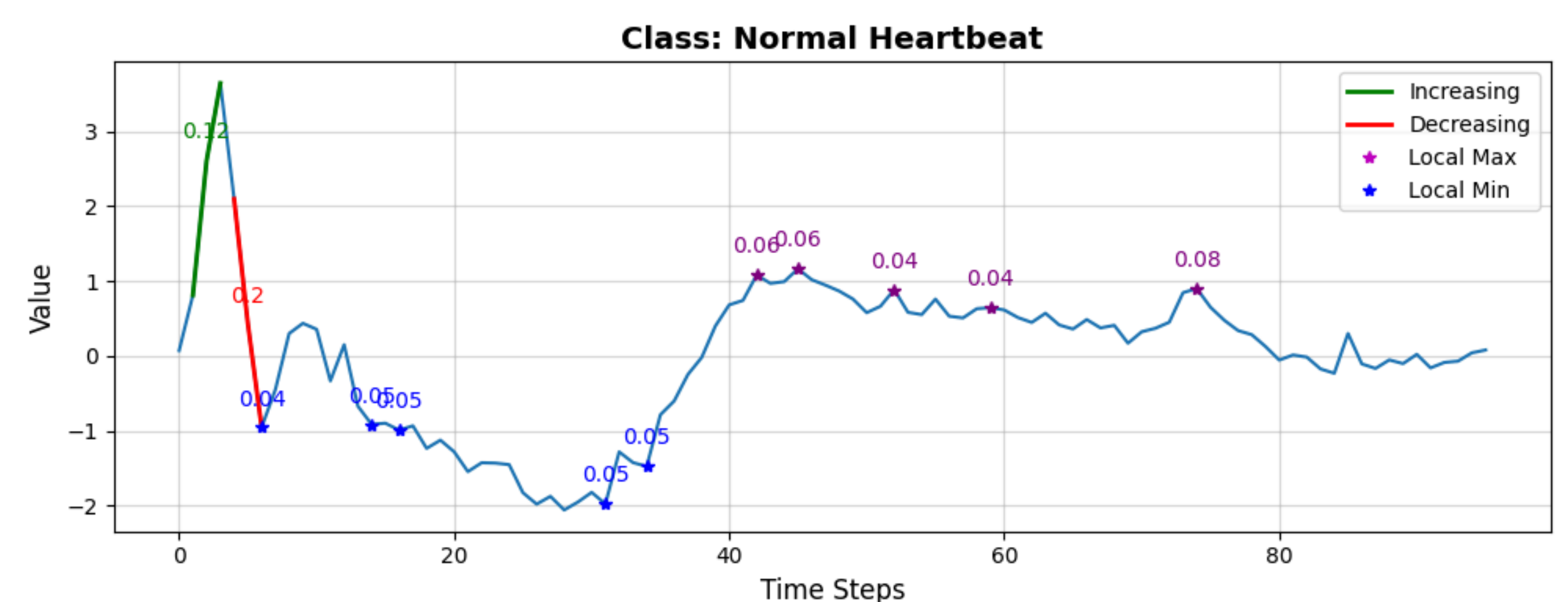


Fig 4: Explanation highlights segment significance, relevance scores, and event types (e.g., increasing, decreasing, maxima, minima).

- Our method highlights each time series segment's significance and relevance scores, detailing event types like increases, decreases, maxima, and minima.
- Perturbation strategies vary across different datasets.

Conclusion and Future works

- Post-hoc XAI methods are being developed to provide local and global explanations for time series classifiers, capturing temporal dependencies and aligning with human interpretation.
- Implement L2GX, extend current methods to multivariate time series, and validate using diverse perturbations across multiple datasets and XAI baselines.

Acknowledgements

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Scan QR codes for papers:



References

- [1] Theissler et al.: Explainable ai for time series classification: A review, taxonomy and research directions. IEEE Access (2022)
- [2] Schlegel et al.: Ts-mule (2021)
- [3] Ribeiro et al.: "Why should i trust you?" (2016)
- [4] Kadous: Learning comprehensible descriptions of TS (1999)